

Navigating Skin Concerns with AI: A Human-Centered Investigation of a Dermatology App in a Diverse Community

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Abstract

AI-powered dermatology apps have the potential to improve access to skin health information, but concerns remain about their accuracy and usability, especially for under-served communities. We contribute a mixed-methods study investigating the impact of an AI-powered skin condition app on a diverse, multilingual population (N=110) in a community setting, involving real skin concerns and in-person consultations with clinicians. We found that app use was associated with a significant increase in participants' ability to correctly name their condition ($p=0.011$). Thematic analysis of interview conversations revealed that users relied heavily on visual matching of example images, highlighting the importance of representative imagery for diverse skin tones and condition severities. Clinicians viewed the app as a valuable tool for facilitating patient-provider communication, acting as a shared reference point. We conclude with considerations around the real-world impact AI tools may have on the quality of care, health equity and patient-provider interactions.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Empirical studies in HCI**; • **Applied computing** → **Consumer health**.

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1 Introduction

Skin concerns are extremely common, with an estimated prevalence of over 2 billion worldwide [15]. Access to care for these common conditions is limited: early estimates suggest only 28% of cases see a dermatologist [13], and wait times range from weeks to months, with longer wait times for under-resourced groups such as people on Medicaid [8].

The difficulties and delays in access motivate the development of tools to help patients better understand and manage their conditions, including artificial intelligence (AI)-based tools. A number of image-based AI models have demonstrated skin condition classification for a range of conditions at or near dermatologist level [12, 17, 25]. Models of this type may assist clinicians such as primary care physicians and nurse practitioners [18], as well as dermatologists [31]. AI tools may also aid lay people in their health-related

decision-making, and consumer-oriented apps are available for detecting common skin cancers, such as SkinVision [29]. Yet many outstanding questions remain around the potential use of AI tools for skin detection by the general public. Inaccurate results may lead to missed treatment and/or may place undue burden on the health care system [29]. There are also questions around the generalization of deep learning models to new populations [16], and whether models perform well on groups under-represented in the training data [9]. Further, existing research may not be assessing the needs of under-resourced communities.

This research aims to better understand, in the context of real-world skin concerns, how an AI-based app affects users' understanding of their concern and decision making, and how it influences subsequent clinician-patient conversations. We engaged 110 community members at a series of public health events in east San Jose, CA, USA. As part of the study, we provided in-person language support for participants speaking Spanish, Vietnamese, and Mandarin, as well as translated versions of the app. Participants used a research prototype of a skin AI app to learn about their skin concerns, and get a set of matching results. Every participant looked over their app results, and then consulted with a clinician, who also reviewed the app results and discussed any concerns with the participant. We adopted a mixed-methods approach to analysis. First, we obtained deep qualitative insights into community members' needs and perceptions around skin concerns before, during, and after their use of the app, and elicited perceptions of clinicians on how the app performs and how it may affect clinical workflows and patient journeys. We also calculated before-after differences in users' perceptions of their concerns. We present qualitative and quantitative insights and discuss them in the context of their potential implications on the quality of care, health equity, and patient-provider communication. In summary, our work makes makes the following key contributions:

- We present insights from a mixed-methods study investigating the impact of an AI-powered skin condition app on a diverse, multilingual population (N=110), involving real skin concerns and in-person consultations with dermatologists.
- We discuss considerations around the real-world impact AI tools may have on the quality of care, health equity and patient-provider communication.

2 Related work

With recent advances in AI methods and availability of larger datasets and compute resources, AI models have been developed for many medical imaging modalities [11]. Among these are applications to dermatological images, with studies showing that models can be trained capable of dermatologist-level performance at classifying dermatological conditions in images [4, 12, 14, 19, 25, 30]. Clinical studies indicate potential benefits in decision support for clinicians, including nurse practitioners, primary care physicians, and specialists [17, 18, 18, 31].

Health information seeking apps for self-diagnosis and triage. There is substantial interest in using AI technology to directly assist laypeople (i.e. non-clinicians) with understanding their skin conditions. A qualitative study on patient perspectives found that patients valued increased diagnostic speed and health care

access, and that these tools, while driving patient activation, could lead to increased patient anxiety [26]. Another study on chatbot-based symptom checkers [34] also found that efficiency of access to information was important for AI-based assistance, but that this desire for speed and access was at odds with a desire by some for more empathetic communication. A Wizard-of-Oz study on a chatbot interface indicated that use of a text-based dermatology assistant could reduce users' level of concerns about their skin conditions [24]. In addition to diagnosis, research has examined specific causes of potential eye concerns, such as specific types of makeup [33]. Smak Gregoor et al [29] conducted a retrospective population-based pragmatic study in the Netherlands, where a large health insurance company offered free access to an assistive dermatology app. Compared to controls who did not use the app, app users filed more claims for skin malignancies, including benign conditions. Babic et al [1] similarly highlight the potentially greater impact of false positives for consumer-facing tools compared to clinician tools. Conversely, a study on the effects of Internet search for health concerns on case vignettes suggested that a relatively small fraction of people changed their assessments, with an overall modest improvement in diagnosis but not triage [23].

Equity and bias in AI for healthcare. The potential widespread use of AI tools for dermatology has raised concerns about the risks of bias and equity impacts. Analyses in other domains have indicated that algorithms used for health care may propagate systemic biases; [27] demonstrated that an algorithm used to assess patient risk assigned the same risk level to Black patients who were sicker than White patients. They estimate that this reduces the number of Black patients identified for extra care by half. [9] identify concerns in the reporting of dermatology AI algorithms, particularly around the reporting of data used to train and validate models. They argue that a lack of transparency in the demographic properties of the data as well as the disease labels used hinders efforts to ensure such algorithms perform well across the population, especially for under-represented groups.

Need for community-centered participatory research on health AI technology. An additional concern around consumer-facing AI technology is the need to actively engage communities that may benefit or otherwise be impacted by the technology [7, 22]. Previous human-computer interaction research on deployment of clinician-facing AI technology has highlighted the importance of considering socio-environmental factors on how AI is utilized in day-to-day use[2]. Studies engaging patient perspectives in other health domains, such as mental health, have indicated that the perceived value of AI tools and their own needs may be misaligned [21, 28] There is also some question as to whether AI tools may be used as a tool that provides information, versus as a collaborator [32]. These considerations motivate early and active engagement with diverse users in the community to assess their perspectives on AI technologies, understand how they may engage with the tools, and inform AI tool design and evaluation.

Our work aims to address these gaps by contributing insights from a mixed-methods study investigating the impact of an AI-powered skin condition app on a diverse, multilingual population in a community setting, involving real skin concerns and in-person consultations with clinicians, and discussing implications for the

real-world impact AI tools may have on the quality of care, health equity and patient-provider communication.

3 Methods

App details. We examined a research-only assistive app (not commercially available), developed to help people learn about their skin concern. The app collects 3 images of a skin concern, and optional metadata questions such as history of skin conditions and how long the current concern has been present. It then returns 3-7 matching results, with each skin condition accompanied by example images and information about the condition (Figure 1).

Setting and recruitment. The study was conducted over a series of 4 community health events in east San Jose, CA, USA from October 2022-May 2023. Events were hosted by Santa Clara Family Health Plan, which serves members of the surrounding community, many of whom rely on a healthcare safety net, Medi-Cal.

All community event attendees aged 18 or older, who spoke English, Spanish, Vietnamese, Mandarin or Tagalog and had a skin concern that they were willing to try the app on were eligible to participate in this study. For all events, participants first used the skin AI app on their concern, and then saw a clinician for an educational screening / consultation (details in Supplementary Methods). For the last 3 events, with institutional review board approval, participants were compensated for their time with a \$15 gift card.

Procedure. The study flow is illustrated in Figure S1A. The first event had a similar but lengthier study flow. Participation involved the following steps: Providing informed consent, completing the pre-app-use survey; using the app while providing feedback to a researcher from the team; completing the post-app survey; consulting with a clinician; and completing the final end-of-study survey asking for perceived satisfaction and helpfulness. Pre-app and post-app surveys probed for participants' level of concern, ability to name their own condition, and intended next step before and after using the app respectively. If applicable (e.g., if the researcher or clinician was not conversant in the participant's preferred language), a volunteer assisted with translations between the participant and researcher/clinician. Further details of study logistics are described in [10].

In addition to surveying and interviewing community participants seeking to understand skin conditions, we also surveyed 9 of the 10 clinicians who provided consultations. During the events, community participants brought the study devices (running the app and showing their results) to clinicians to facilitate discussions. Clinicians then completed one post-consultation survey for each participant, indicating whether the app results were consistent with their own impressions of the case, and their perceptions of the interaction. Over the course of the study, clinicians also completed semi-structured interviews with researchers, to provide overall impressions of the app, and what role it played or could play in clinician-patient interactions.

Ethical Considerations & Positionality Statement. This study was conducted by researchers affiliated with Google and Stanford, and staff at the Santa Clara Family Health Plan. The study protocol was reviewed and approved by two institutional review boards

(IRB): Stanford (covering activities by Stanford researchers) and Advarra (covering activities by Google researchers).

Data analysis. Quantitative data analyses were performed using Python 3.0 and the SciPy package. Pre-app vs. post-app comparisons used the McNemar test for binary values (such as accuracy), and the Wilcoxon signed-rank test for ordinal responses. 95% confidence intervals on all metrics are reported using bootstrapping with 1,000 bootstrap samples. Condition-naming accuracy was measured against a panel of 3 US board-certified dermatologists per case [18, 25]. We applied thematic analysis [3] to qualitatively analyze participants' and dermatologists' survey responses. Themes were inferred inductively until theme saturation. Two researchers were involved in the data analysis and synthesis process.

4 Results

A total of 133 community members participated in the study (Figure S1B). Of these, 19 were consented but withdrew from the study before using the app (e.g., they had a skin condition that comes and goes, and it was not present on that day), and were excluded from further analysis. Of the remaining 114, 2 withdrew after using the app (i.e., ran out of time and/or declined to see the clinician), and the remaining 112 completed the study. For two participants, the app withheld predictions, based on an app function that detected potentially out-of-distribution cases (i.e., cases for which the AI's confidence was not sufficiently high); the remaining 110 were used to evaluate app accuracy. Overall accuracy of the algorithm on this population was 83%, as assessed by the inclusion of a dermatologist panels' top condition being included in the result set (more details in Appendix).

Overall app perceptions. Overall, participants overall viewed the app as helpful: 72% very or extremely satisfied with the app; and 64% very or extremely helpful at helping decide on next steps for their concern (Table 1). Participants felt the app could help them be more prepared for discussions with clinicians ("Was helpful in the interaction as it served as a record of the condition," P46; "I think it helps. Because I can show the doctor why I think it is a specific syndrome. Helps with making communication easier," P47; "Yes, it helps make the conversation easier," P94). Clinicians viewed the app's results as mostly consistent with their own evaluations (86% consistent or very consistent), and that it was a helpful tool in facilitating conversations with participants (92% helpful or very helpful; Table 2). Specifically, clinicians were able to use the app to anchor discussions with the participants, and use it to explain possible matching conditions, expand on details about the conditions, or frame discussions based on patient concerns. Clinicians also appreciated the example images of each condition, but reported a preference for greater customization, such as for disease severity, to avoid undue concern on the patient's part. We provide additional qualitative insights from discussions with clinicians in Supplementary Section C.

4.1 User perceptions of app

4.1.1 Desired information from app. Before using the app, participants expressed a wide range of desired information. When asked

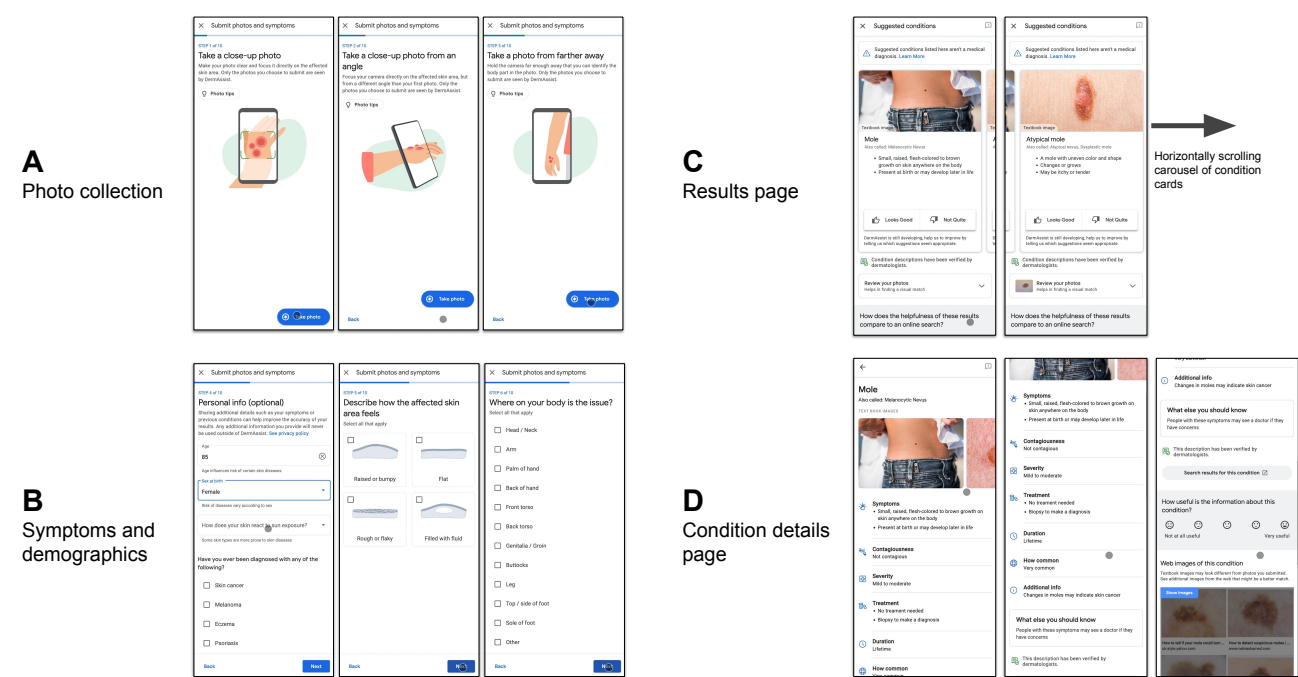


Figure 1: Illustration of app flow. Screenshots of English-language version of app showing example views of app during the case submission flow (A and B) and the results containing matching conditions (C and D).

what they would expect the app to show them, participants commonly expressed a desire for treatment information, both to address their specific concern (“Expected to get information about the treatment of the problem”, P38; “Treatment info for spider veins”, P60), as well as proactive treatment steps to prevent recurrence (“Follow-up actions: Maybe apply more sunscreen or wash my face?”, P26).

Participants also wanted to decide if they needed to seek additional medical care. In some cases, they expressed a desire to identify a specific condition (“Expects to see a diagnosis result coming up, and participant anticipates seeing a result related to freckles due to aging”, P15). But more often, the expressed need centered around the overall seriousness of the concern, without expecting to identify a single matching condition (“Expect an idea of what it could be rather than diagnosis”, P33; “Diagnosis of what this is; possible cases. Also: what SEVERITY is this. Do I need to see a doctor now? Or when?”, P55; “What the conditions are and if they should see a doctor,” P81). In some cases, the concern was about classes of conditions, or about progression: for instance, when asked what they would hope to learn from the app, P109 responded “Is it cancer?”, while P67 responded that they “would expect to see how bad the condition is aka how far it has progressed”.

4.1.2 Submitting skin concern information. Participants did not report substantial difficulty in inputting details of their concerns into the app; though this was likely affected by the study’s setup, such as having researchers available to assist with taking images of their skin concern when difficult to photograph on their own. A common challenge was obtaining images with sufficiently high

quality using camera phones; the app’s internal image quality check required ensuring features were in focus, resulting in some participants requiring several attempts to obtain a high-quality image. Most participants seemed comfortable with optional metadata questions, though a few expressed concern or confusion about why they were being asked some questions. (“Sun exposure – why is it asking this”, P72.) During the final event, the dynamic case submission process reduced the number of questions needed on average (Methods; Table S3); compared to 76 participants in prior events who were always asked 10 questions, 16 / 34 participants in the last event were asked no questions after image capture. However, for 6 / 34 participants, the dynamic case submission model asked a large number of questions (11 - 20); every participant in this group commented that this felt like too many questions.

4.1.3 Understanding app results. When reviewing app results, participants strongly relied on visually matching their skin concern to example images. Participants wanted to see example images matching body part and skin tone to their own concern (“The body location is a little off. like it shows the hair part when the participant was looking for the facial problem,” P31; “First textbook [example] image showed a foot even though she selected ‘palm of hand’, P56). Participants also assessed the severity of the example images in relation to their own to determine if a condition might be a good match (“That’s not my picture. My problem is better than that,” P17). For some cases, the example images showed a strong mismatch in context or apparent severity to the participants’ impressions of their own condition. For instance, P82 saw a results card for “scar

Response	Users	%
How satisfied are you with the app you just used?		
Not at all satisfied	2	2.2%
Slightly satisfied	2	2.2%
Moderately satisfied	21	23.1%
Very satisfied	33	36.3%
Extremely satisfied	33	36.3%
(No response)	19	
Total	91	100.0%
Did you view [the app] as helpful in determining what steps you should take next for your condition?		
Not at all helpful	2	2.2%
Slightly helpful	8	8.8%
Moderately helpful	22	24.2%
Very helpful	31	34.1%
Extremely helpful	28	30.8%
(No response)	19	
Total	91	100.0%

Table 1: User perceptions per post-app responses.

Response	Clinicians	%
To what degree did you view the suggested conditions from [the app] as consistent with your own assessment?		
Very inconsistent	4	4.3%
Inconsistent	9	9.6%
Consistent	39	41.5%
Very consistent	42	44.7%
(No response)	16	
Total	94	100.0%
Was [the app] a helpful tool in the conversation with the participant?		
Unhelpful	7	8.4%
Helpful	49	59.0%
Very helpful	27	32.5%
(No response)	27	
Total	83	100.0%

Table 2: Clinician perceptions per post-consultation responses.

condition”; but the example image showed a large hypertrophic scar that was much larger than the participants’ own concern. This likely increased the participants’ own self-reported level of concern.

Parsing a long list of results from the app was sometimes a challenge. For 56% of participants, only 3 result cards were shown; but for 35% of participants, the app presented 7 result cards. This could feel like too many, and sometimes feel difficult to navigate

(“There are a lot of possibilities. not necessarily a good thing, can be confusing because I don’t think I have those”, P62). While conditions were presented in order of highest strength of match, this was not always apparent to participants, several of whom expressed a desire to see confidence indicators for each result card.

Exploration of results largely centered around horizontally-scrolling results cards on the main results page. It was less common for participants to explore the details page for each specific condition, or to see the extended list of conditions. Only 22% of participants discovered the condition details page without any prompting from the researcher; while 49% discovered it with a gentle prompt. The remaining participants needed more explicit prompting to discover the condition details page. When exploring the details, most participants commented that the information was useful and comprehensible; common treatments for a condition were most commonly cited as the most relevant information. Some participants expressed a desire to connect to providers from the condition details page (“It should create a link you can share with your doctor or telehealth service”, P67).

Increased success in identifying appropriate matching conditions. Compared to before app use, after using the app participants were more likely to both respond they were comfortable naming a condition for their skin finding (49.3% vs. 29.6%, $p=0.003$), and to name a condition matching dermatologist diagnoses (15.5% vs. 6.0%, $p=0.01$; see Table 3). Anecdotally, reasons given for not naming the condition include believing that to be a task for doctors and not having time (within the span of this study) to do more research. Reasons for their provided condition names not matching conditions in the reference differential diagnoses were most typically due to naming non-specific conditions or symptoms such as rash, allergy, dry/itchy skin, spots, and moles.

Impact of app use on level of concern and planned next steps. The overall distribution of participant-reported level of concern and planned next step did not change significantly after app use (Table 3). To better contextualize these shifts, three dermatologists reviewed these changes. Among 39 cases with changes in self-reported level of concern, 30 (76%) were more consistent with the dermatologist panel differential; among 32 changes in planned next step, 24 (75%) were more consistent. The tendency towards greater consistency was more pronounced when participants’ level of concern or next step severity decreased after use (85% vs 58% more consistent for level of concern decrease vs increase; and 83% vs 64% for next step) (Table 4).

5 Discussion

This study engaged a diverse participant pool, speaking a variety of languages, to provide their perspectives on an AI-powered dermatology app after using the app on their skin concerns. We found that the app remained accurate in this population, and collected qualitative feedback and quantitative data on the effect of app use on user sentiments. Compared to before app use, participants were more comfortable guessing their skin condition’s name post app use, and had higher accuracy guesses. Participants tended not to change their level of concern or planned next step for care, but

Metric	Pre-app use (Percent, [CI])	Post-app use (Percent, [CI])	N	p
Felt comfortable guessing condition name	31.6 [22.4, 41.8]	43.9 [34.7, 54.1]	98	0.028
Guessed condition name matching dermatologist differential	6.0 [1.2, 11.9]	15.5 [8.3, 23.8]	84	0.011
Very/Extremely concerned	37.8 [30.5, 48.8]	29.3 [20.7, 39.0]	82	0.145
Would see doctor next step	17.9 [10.7, 25.0]	15.5 [08.3, 22.7]	84	0.677
Very/Extremely confident in next step	61.4 [48.6, 72.9]	68.6 [57.8, 80.8]	70	0.354

Table 3: Summary of pre-app use / post-app use comparisons. Each row reflects a different measurement of participant responses to survey questions before and after app usage. N represents the number of participants who answered a given question both before and after app usage; because questions were optional, not all participants answered these questions. p values reflect non-equivalence tests between pre and post app distributions; binarized measures like accuracy were compared with the McNemar test, while ordinal measures like scale responses were compared with the Wilcoxon signed-rank test.

	Level of concern			Next step severity		
Dermatologist review of changes: Is the change more or less consistent with the differential?	Decrease	Increase	Unchanged	Decrease	Increase	Unchanged
Less consistent	4	5	50	3	5	41
More consistent	23	7		15	9	

Table 4: Dermatologist manual review of changes in participant-reported level of concern and planned next step for their skin concerns .

changes were determined by dermatologists to be most often consistent with the assessment of a separate panel of dermatologists. Presenting example images matching participants' own concerns was a key usability need.

Implications for real-world impacts of AI tools. Previous work has indicated potential benefits and risks with consumer-oriented AI health tools: They may extend access to high-quality care, but also lead to systematic over- or under-diagnosis [1, 5, 29]. Further, such tools may amplify systematic inequities by providing higher-quality information for some subgroups compared to others, or providing biased information [9, 27]. Our findings further the understanding of these issues by providing more data on app performance generalization to a new and diverse cohort and showing that AI app use can substantially increase success at finding a correct matching condition. Participants' success rate at identifying their condition with a name that matched dermatologist assessment increased after app use; although there is substantial room for further improvement.

Implications for patient-provider interactions. In addition to user impacts, the impact of AI apps on providers will be important to understand. Clinicians in our study indicated substantial potential

for the app to be a valuable communication tool to facilitate patient-provider conversations (Table 2). App predictions were viewed as largely concordant with provider assessments; but situating the information about matching conditions in a conversation may be the best modality for helping people understand the information presented. This is potentially consistent with qualitative research on patient perspectives, suggesting that patients view the potential of AI tools in terms of the human physician-patient relationship [26]. It also suggests that informational apps such the one studied here may serve as boundary objects, providing a common frame of reference between layperson and clinician understanding, similar to how onboarding instructions have been reported to align clinical and AI developer needs and expectations [6].

Important aspects of the user experience. Our sessions identified two aspects of user needs for which the app could improve. First, the images shown for conditions were originally intended to showcase a small number of representative examples of the skin condition. However, mismatches between the example images and participants' own images posed difficulties for participants identifying matching conditions, and led in some cases to incorrect assessments. Example images should ideally match the body part, severity, and skin type of participants to help with visual pattern

matching, with clearer affordances to see additional examples. Second, participants expressed a strong desire for more information about AI confidence. While conditions were presented in order of confidence in the app, there were no explicit signals about degree of confidence. More explicit indicators may help; however, we caution that providing AI confidence is not straightforward. For one, the mathematical probabilities output by AI models may not be well calibrated or properly account for priors and disease severity, and for another, helping people understand probabilities in a uniform way is hard, a challenge similar to that in meteorology and peoples' understanding of the probability of precipitation [20].

Limitations. The present work has limitations. While this was a large qualitative study, it was not a large geographically distributed study. As might be expected for the sample size of 110 participants, there were relatively few cases indicated by the dermatologist panel to be potentially serious or rare conditions; this limited our ability to reliably quantify impacts for such rare or serious cases. Since all participants were able to consult with a clinician immediately, we could not also observe what downstream follow-up behavior might have been for people without immediate access, or participants' sentiments if they had more time (hours or days instead of minutes) to explore the app and do additional research. Furthermore, since the study setup involved helping participants if needed during app use, like taking images of the concern, we were unable to assess potential difficulties people might encounter in these stages of app use without such assistance.

6 Conclusion

In this work, we contributed insights from a mixed-methods study investigating the impact of an AI-powered skin condition app on a diverse, multilingual population in a community setting, involving real skin concerns and in-person consultations with dermatologists. We discussed considerations around the real-world impact AI tools may have on the quality of care, health equity and patient-provider communication, and encourage the research community to explore these aspects in other community settings, clinical domains and interaction modalities.

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A Supplementary Methods

A.1 Translated versions of the app

The dermatology app text was translated from English into four additional languages for the study: Mandarin, Spanish, Tagalog, and Vietnamese. These languages were selected based on their frequency of use by patients in the community health service partners for the study. All user interface text was translated by Google staff fluent in the respective language; while clinical text, such as descriptions of skin conditions, was translated by clinicians fluent in the respective languages. Screenshots of the app in each of the translated language are shown in Figure S2.

In addition to using translated versions of the app, participants who primarily spoke a language other than English had language support for each session. This language support was provided by one of the researchers running the session, if they were conversant in the language; or by dedicated volunteers fluent in each language.

The main study had participants using the app in English, Mandarin, Vietnamese and Spanish. Although some community members speaking Tagalog participated in the clinician-consult stage of the study, they did not complete all steps of the study and thus were not included in the results presented here.

A.2 Data management

All study data were de-identified. Participants were assigned unique IDs (e.g. “SD001”, “SD002”, ...), and these IDs were used to pseudonymously join each user’s responses to different surveys, researcher notes, app-provided predictions and dermatologist assessments of each case.

A.3 Statistical analysis for supplemental results

All analyses were performed using Python 3.0 and the SciPy package (<https://scipy.org/>). Pre-app vs. post-app comparisons between binarized variables, such as accuracy measurements, used the McNemar test. Pre/post comparisons between ordinal values, such as Likert survey responses, used the Wilcoxon signed-rank test. 95% confidence intervals on all metrics are reported using bootstrapping with 1,000 bootstrap samples.

Accuracy assessments used the reference differential diagnosis (a ranked list of possible diagnoses) from a panel of 3 US board-certified dermatologists who reviewed each case, using methods similar to those described in [18, 25]. AI prediction accuracy metrics were computed as follows: top-n accuracy quantifies whether the top n conditions predicted by the model include the top reference diagnosis from the dermatologist panel. For “top-k” predictions, the number of conditions is variable, depending on the number of conditions surfaced in the app UI for a given case (and based on the confidence of each prediction). Differential diagnoses could be obtained from the panel for the vast majority of cases (103/110 participants). This analysis excludes model predictions for “typical skin”, since that output may accurately describe some cases, but would not be included in the dermatologist differential. As a result, for top-1 accuracy in particular, cases with a top prediction of “typical skin” are excluded from analysis.

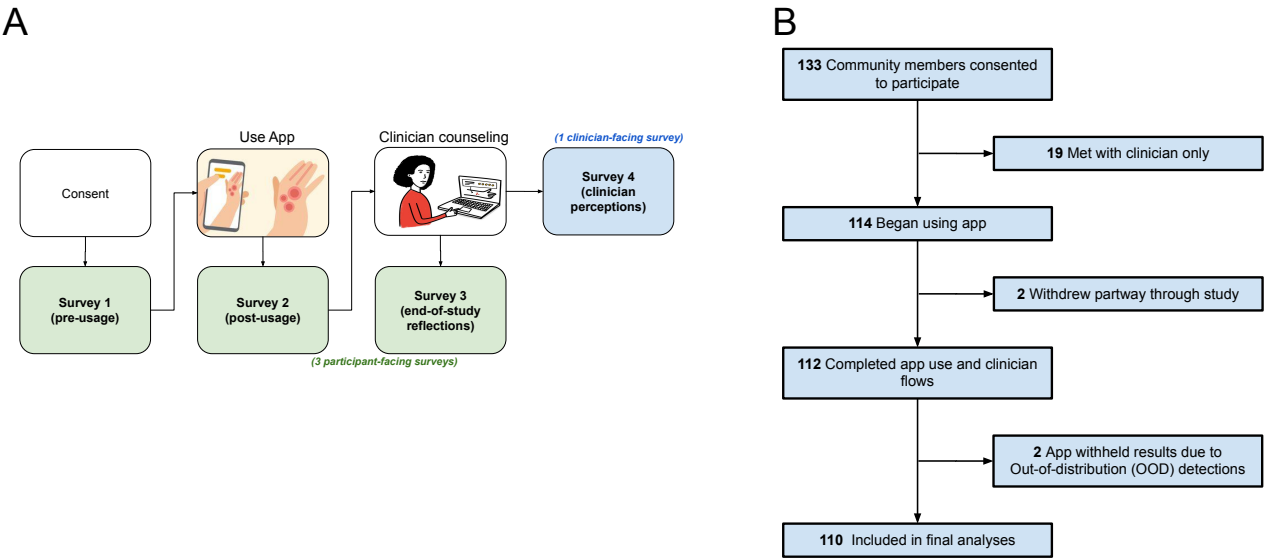


Figure S1: Study design and participant diagram. A, schematic of study flow. B, chart of participant flow aggregated across all events.

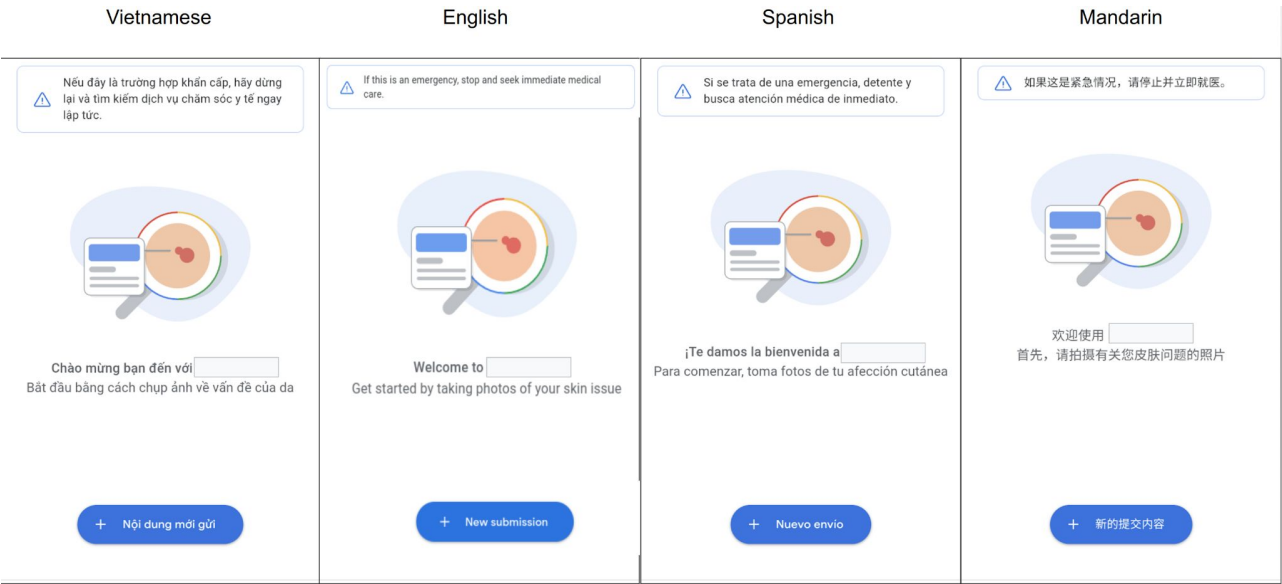


Figure S2: Multiple-language screenshots of the app. The app home page / upload screen is shown in Vietnamese, English, Spanish and Mandarin. The app name is covered with a box for anonymization purposes.

Table S1: Summary of app prediction accuracy compared against the differential diagnosis produced by a board-certified dermatologist panel. “K” in top-k indicates the number of matching conditions shown in results for each participant; k varies from 3 to 7. Values are computed on the 103 cases for which a differential diagnosis was determined by a dermatologist panel. The lower N for top-1 values reflect the exclusion of cases for which the top result was “Typical skin;” these cases are excluded from this analysis because dermatologist differentials typically do not include this.

Result set	Accuracy	95% Confidence Interval	N
Top k	0.83	[0.75, 0.89]	103
Top 1	0.52	[0.41, 0.62]	87
Top 2	0.73	[0.64, 0.82]	103
Top 3	0.80	[0.72, 0.86]	103
Top 4	0.82	[0.74, 0.88]	103
Top 5	0.82	[0.74, 0.88]	103
Top 6	0.83	[0.75, 0.89]	103
Top 7	0.83	[0.75, 0.89]	103

B Supplementary Results

B.1 Prediction accuracy against reference diagnoses

We assessed the accuracy of predictions made by the app relative to differential diagnoses produced by a dermatologist panel reviewing each case. Board-certified US-based dermatologists reviewed all images and case details for each of the cases in the study, providing a differential diagnosis: A ranked list of the most likely conditions matching each participant’s concern. These differentials were compared to the sets of matched conditions provided by the AI dermatology app using metrics similar to those in the literature [18, 25].

Accuracy of the AI app results against dermatologist reference diagnoses is summarized in Table S1. Overall accuracy of the top k conditions against the top reference diagnosis from the dermatologist panel is 82.5%. “k” reflects the variable number (between 3 and 7) of conditions shown for a case; see Table S3. The top-3 accuracy (examining just the first 3 results) is 79.6%. These values are comparable to the previously reported performance of a deep learning system on a retrospective test set of tele-dermatology cases [25], and higher than reported non-specialist performance with or without AI assistance [18, 25].

B.2 Further analyses of next step and level of concern changes

Since the appropriate next step depends on the underlying concern, we examined per-participant shifts in planned next step and level of concern (Figure S3). Among 48% of participants who answered both pre- and post-app survey questions, the reported level of concern did not change. Among the remaining 52% whose level of concern changed, more participants reported decreases (36%) versus increases (16%). For planned next steps, most participants (60%) did not change their intended next step with app use; while the

remaining were split between those whose next step became more urgent (23%) or less urgent (17%). These additional observations motivated the review of changes with dermatologists reported in the main results in Table 4.

C Additional Qualitative Results

C.1 Clinician perceptions of app

Clinicians reported that the app’s results were mostly consistent with their own evaluations, and that it was a helpful tool in facilitating conversations with participants. Across participants, clinicians reported that the app predictions were most often consistent (41%) or very consistent (45%) with their own assessments; and that the app was most often helpful (59%) or very helpful (33%).

C.1.1 Use of app to facilitate conversations. Clinicians reported that the app often facilitated discussions with participants about their conditions. Clinicians would anchor discussions around information displayed by the app. They also discussed ways in which the app may be used for future interactions with their patients, as a way for patients to communicate concerns with them. Aspects in which clinicians saw value in the app to anchor conversations included:

- Explaining a set of different possible matching conditions (C2: “The options were spot on and helped clinically reinforce the things that I was talking about with the patient.”)
- Providing a record of details of specific conditions. For instance, C5 reported a participant who may have had tinea versicolor, which was visibly noticeable, but was not concerning in severity. The condition information in the app helped reassure the patient, supporting the clinician’s reassurances.
- Initial framing of discussions based on patient concerns. (C1: “It would speed up some conversations actually, because it could allow me to understand their level of concern and everything coming in. And what information have they already consumed, that may be shaping their mindset coming in. So I could use those as anchors for my conversation with the patient”, C5: “I would love a message saying ‘I have this spot on my arm, [the app] said it could be a basal cell skin cancer. I need to see you.’ So it weeds out a lot of the worried-wells and then also a lot of the patients who don’t worry enough”)

C.1.2 Example images. Diversity of example images was appreciated by clinicians. Once discovered, the additional example images provided in the condition card were noted to be very helpful, as they increased the likelihood that the patient could find an image which matched their presentation. At the same time, clinicians also noted several areas of potential adjustment to the images selected. In particular, they noted that the presentation of certain conditions varies by body part, skin tone, patient age and condition severity and suggested that users of the app could be better supported if the images presented in the results overview were selected to match their personal characteristics based on photo upload and metadata provided. For example, C7 elaborated: “So the diagnosis was correct, the app diagnosed eczema, but the first multiple images they showed her were really severe cases of atopic dermatitis. So she

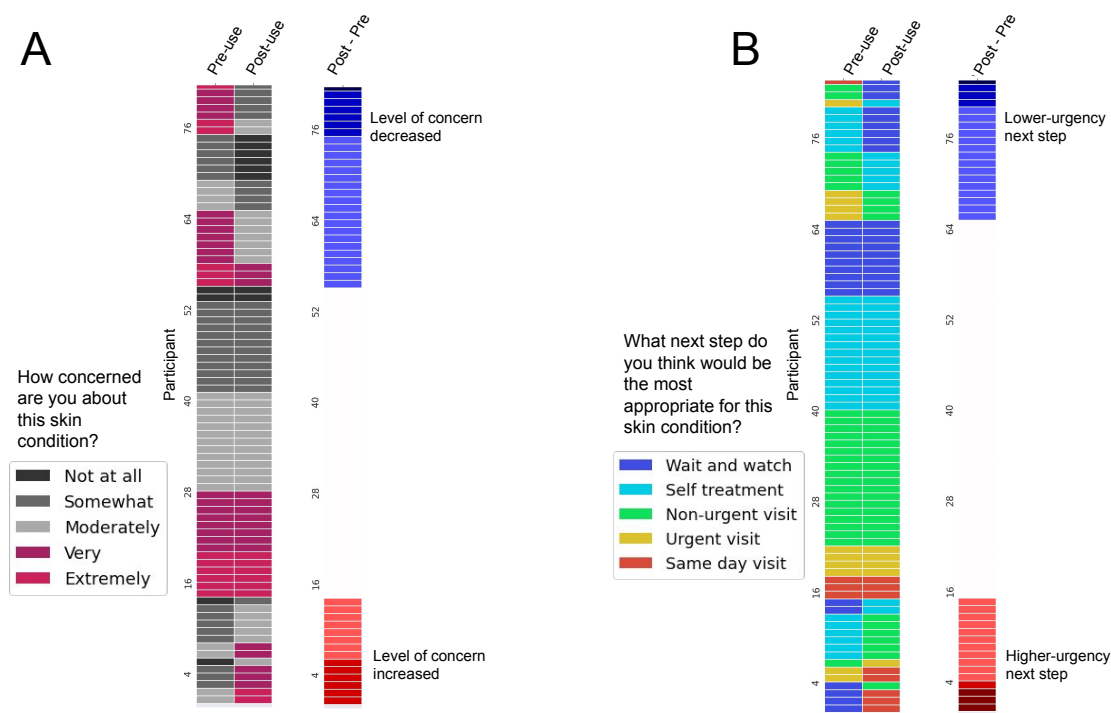


Figure S3: Changes in participants' level of concern (A) and planned next steps (B).

kind of freaked out she said, 'I saw these pictures and I'm thinking- am I going to get that bad?' "

C.1.3 Conveying urgency and likelihood. Beyond condition name, example images and metadata, clinicians emphasized the importance of conveying the urgency and likelihood of each condition in the results overview, suggesting that this information will help the patient to take the right next step or reassure them that no action is needed. To this end, clinicians thought it would be helpful if the app displayed probabilities for the conditions listed to support users in deciding how likely it is that they should make an appointment with their doctor. For example: "That's kind of the question, right? Is this a mole or is this melanoma? Is this something that I see my dermatologist at my next routine? Or do I call and make an appointment today? So have a probability, have the [the app] give the differential. Maybe like: 75% or 95% likely it's a regular mole but it could be melanoma- look for these features.... would be helpful for patients to figure out how urgent this is" (C5). Clinicians also noted that patients are reassured by knowing what category a given condition belongs to, especially for categories that indicate urgency (e.g., contagious, cancerous).

Breakdown	Participants	%
No. matching conditions		
3	62	56.4
4	1	0.9
5	5	4.6
6	3	2.7
7	39	35.5
Total	110	100.0
No. metadata questions asked		
0	16	14.6
1	7	6.4
2	3	2.7
3	1	0.9
8	1	0.9
10	76	69.1
11	1	0.9
12	1	0.9
15	1	0.9
19	2	1.8
20	1	0.9
Total	110	100.0

Table S3: Summary of app case characteristics. ‘No. matching conditions’ refers to the number of conditions predicted with high confidence by the app and surfaced in the main result set; an additional extended set of results was available in the app through a link at the end of the results carousel. ‘No. metadata questions asked’ reflects the number of optional questions asked of users around the skin condition. The values for 10 questions asked reflects the first 3 study events, where the same set of questions were always asked. Other values reflect the last event, where a dynamic case submission flow was used that assessed whether the AI model had enough information to confidently assess results given the current information (see [16]; none of the participants in the last event were asked exactly 10 questions). More than 10 questions were asked of some participants in the last event because the dynamic submission flow broke up some questions into shorter, simpler ones.

D Additional Supplemental Tables

Breakdown	Participants	%
Language		
Vietnamese	34	30.9
English	33	30.0
Spanish	30	27.3
Mandarin	9	8.2
n/a	4	3.6
Total	110	100.0
Race/Ethnicity		
Asian	53	48.2
Black or African American	1	0.9
Hispanic, Latino or Spanish Origin	17	15.5
Native Hawaiian or other Pacific Islander	2	1.8
White	1	0.9
Prefer not to answer	3	2.7
n/a	36	32.7
Total	110	100.0
Comfort filling medical forms		
Not at all confident	2	1.8
Slightly confident	10	9.1
Moderately confident	14	12.7
Very confident	31	28.2
Extremely confident	13	11.8
n/a	40	36.4
Total	110	100.0
Comfort using mobile apps		
Not at all confident	9	8.2
Slightly confident	14	12.7
Moderately confident	13	11.8
Very confident	25	22.7
Extremely confident	8	7.3
n/a	41	37.3
Total	110	100.0

Table S2: Breakdowns of study participants characteristics. n/a indicates that a survey question was omitted (not asked in the first study event) or that the participant did not answer the survey question. Breakdown of 110 participants who completed all parts of the study, and received predictions from the app.

Condition	Count	Mean rank
Eczema	41	1.9
Seborrheic keratosis	26	1.9
Dry skin	24	2.2
Sun spot	23	1.6
Psoriasis	14	2.8
Mole	12	1.8
Irritant contact dermatitis	11	3.2
Melasma	8	2.1
Skin allergy	7	3.3
Precancer of the skin	6	2.2
Dandruff	6	1.8
Post inflammatory hyperpigmentation	6	3.3
Skin tag	5	1.6
Neurodermatitis	5	2.2
Squamous cell carcinoma	4	3.3
Insect bite	4	2.0
Acne	4	1.3
Pigmented purpuric eruption	3	2.3
Drug rash	3	2.0
Rosacea	3	2.3
Wart	3	3.3
Male or female pattern baldness	3	1.7
Stasis dermatitis	3	3.3
Cyst	3	2.3
Telogen effluvium	3	3.0
Hemangioma	3	1.3
Lichenification	3	2.0
Lichen planus	3	3.3
Melanoma	3	3.0
Hypersensitivity	3	4.0
Tinea versicolor	3	3.0

Table S4: Ranked list of conditions in dermatologist differentials for cases in the study. List is limited to conditions that appeared 3 or more times.

Condition	Count	Mean rank
Eczema	55	2.3
Typical skin	49	2.7
Sun spot	41	2.3
Mole	33	2.4
Seborrheic keratosis	31	2.6
Dry skin	26	2.6
Psoriasis	24	4.2
Skin allergy	23	5.1
Dandruff	14	3.4
Precancer of the skin	11	3.5
Post-inflammatory hyperpigmentation	11	3.7
Irritant contact dermatitis	10	4.3
Acne	10	2.8
Atypical mole	10	2.6
Male or female pattern baldness	9	2.8
Melasma	9	2.0
Folliculitis	8	3.1
Telogen effluvium	7	3.1
Rosacea	7	4.1
Melanoma	6	2.7
Scar	6	2.7
Ringworm	6	3.7
Neurodermatitis	6	4.3
Hives	4	3.3
Dermatofibroma	4	2.8
Bruise	3	3.3
Insect bite	3	4.7
Basal cell carcinoma	3	6.7
Tinea versicolor	3	4.0
Lichen planus	3	6.0
Squamous cell carcinoma	3	3.0
Inflicted skin lesions	3	4.3
Vitiligo	3	2.7
Skin tag	3	3.0
Knuckle pads	3	5.0

Table S5: Ranked list of conditions in app predictions. Note that for some conditions, the app condition card displayed a common name for the condition (e.g., “Sun spot” for “Lentigo”), with the dermatological name displayed secondarily (E.g., “Also known as: Lentigo”). List is limited to conditions that appeared 3 or more times.